PIPM: Partial and Incremental Page Migration for Multi-host CXL Disaggregated Shared Memory

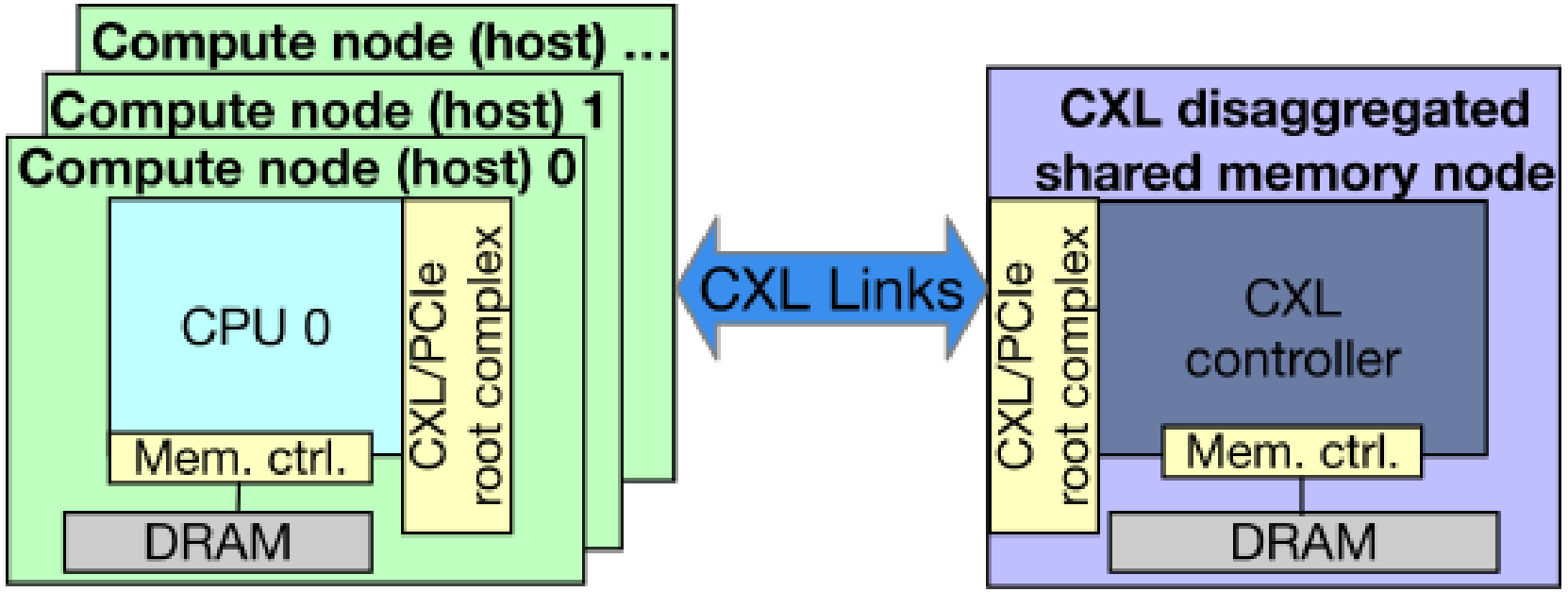
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Background : CXL-DSM

- CXL-DSM : CXL with multi-host, cache coherent, disaggregated shared memory



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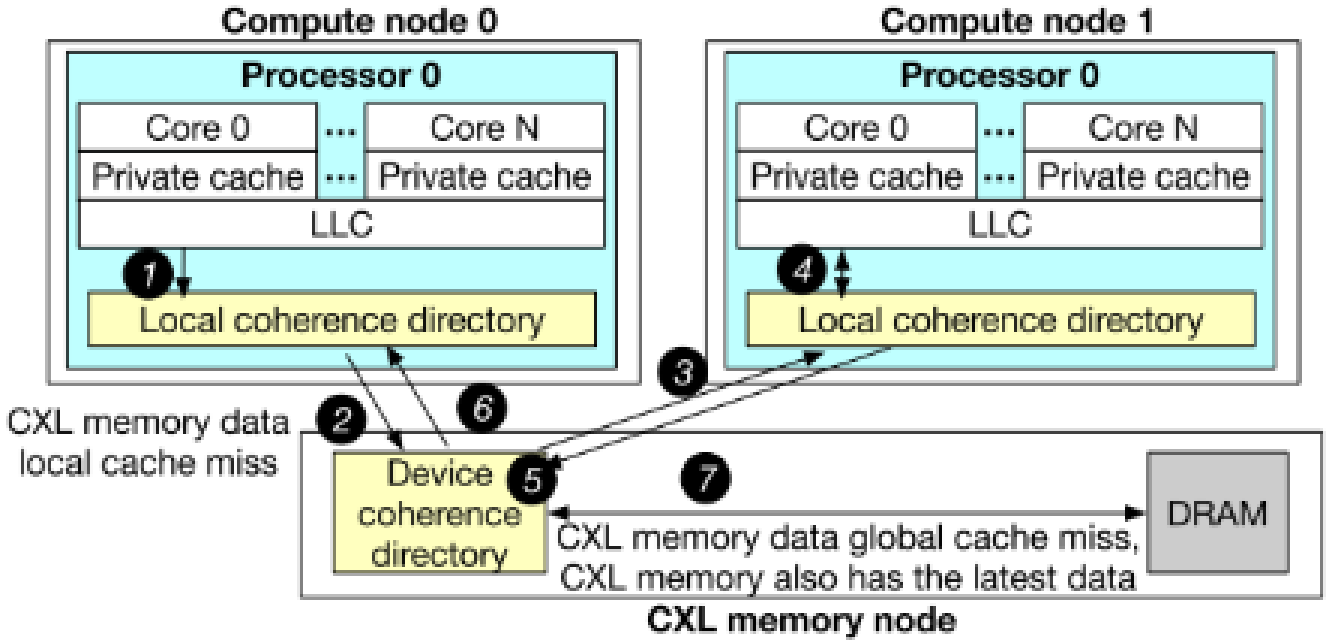


Figure 2. Coherence design of CXL-DSM.

Background : Page Migration

- Limitation of CXL-DSM : high latency of remote access (x2,3 slower than local DRAM access)

=> page migration :

migrate frequently accessed page to host's local DRAM

corresponding migrated page is non-cacheable for other hosts

Existing Page Migration Schemes

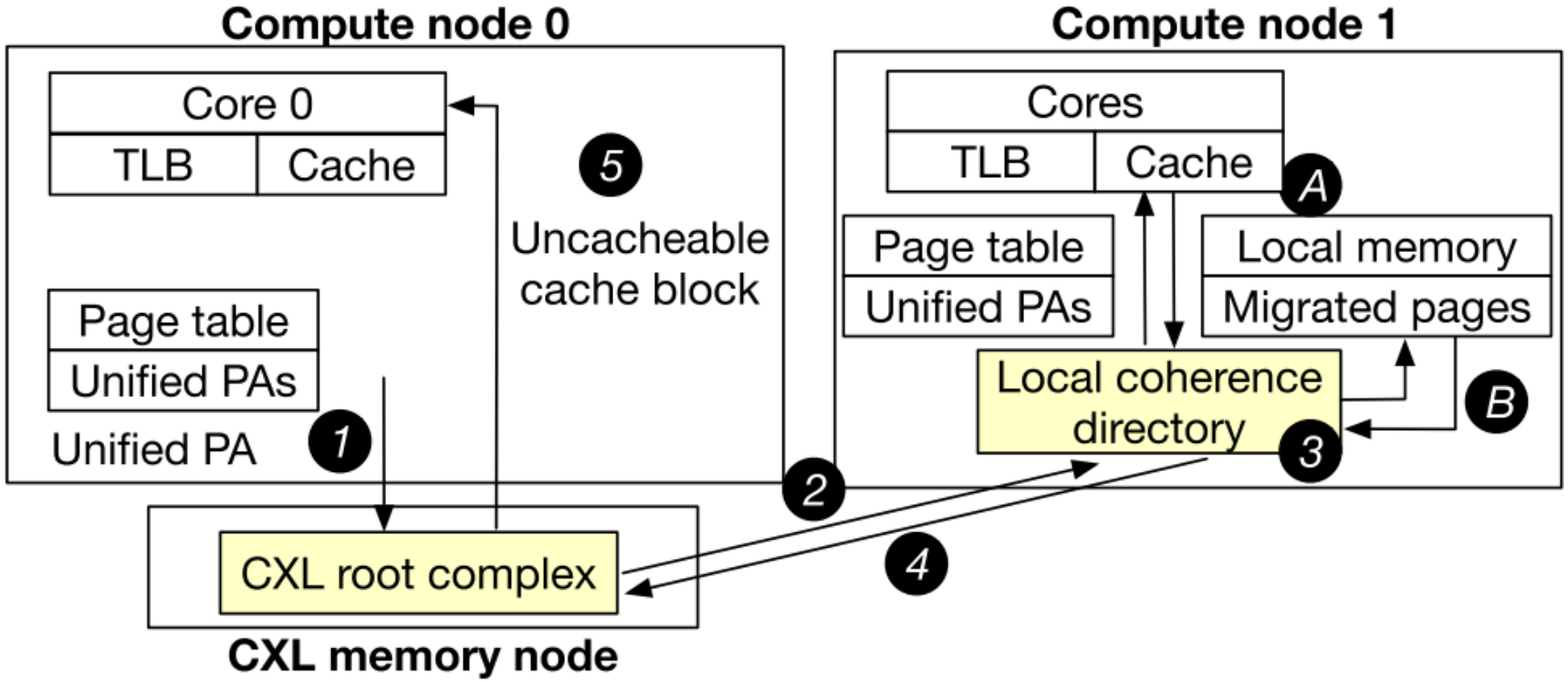
- Problem : designed considering only single-host CXL-DSM

=> ineffective in multi-host CXL-DSM

Prob1 : Local gain global pain

Prob2 : Poor migration scalability

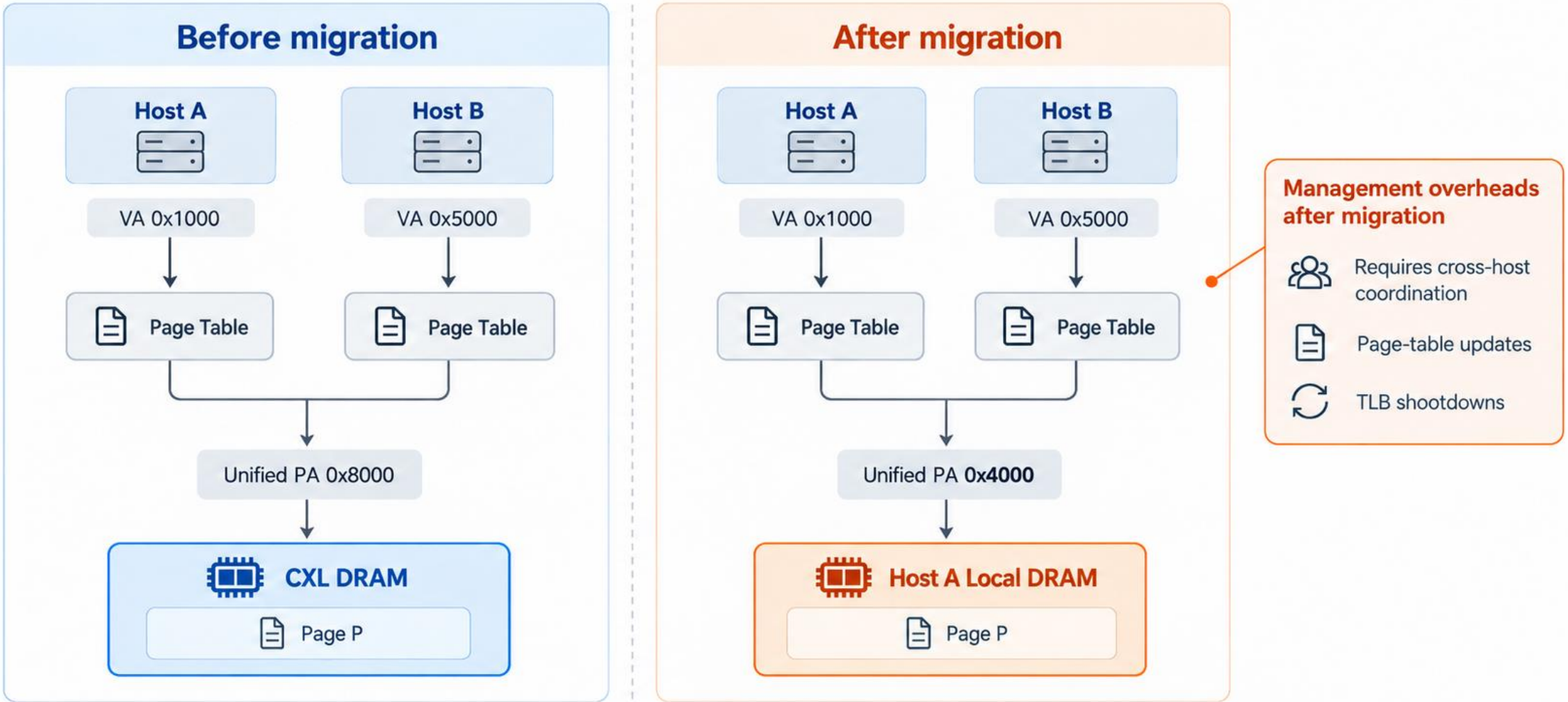
Local gain global pain



Inter host access : migrated page is non-cacheable => 4-hops required

Increase latency to other hosts

Poor migration scalability



Whole-page migration in multi-host CXL-DSM incurs extra coordination and address-management overhead.

Poor migration scalability

- Prior works' assumption : long migration intervals

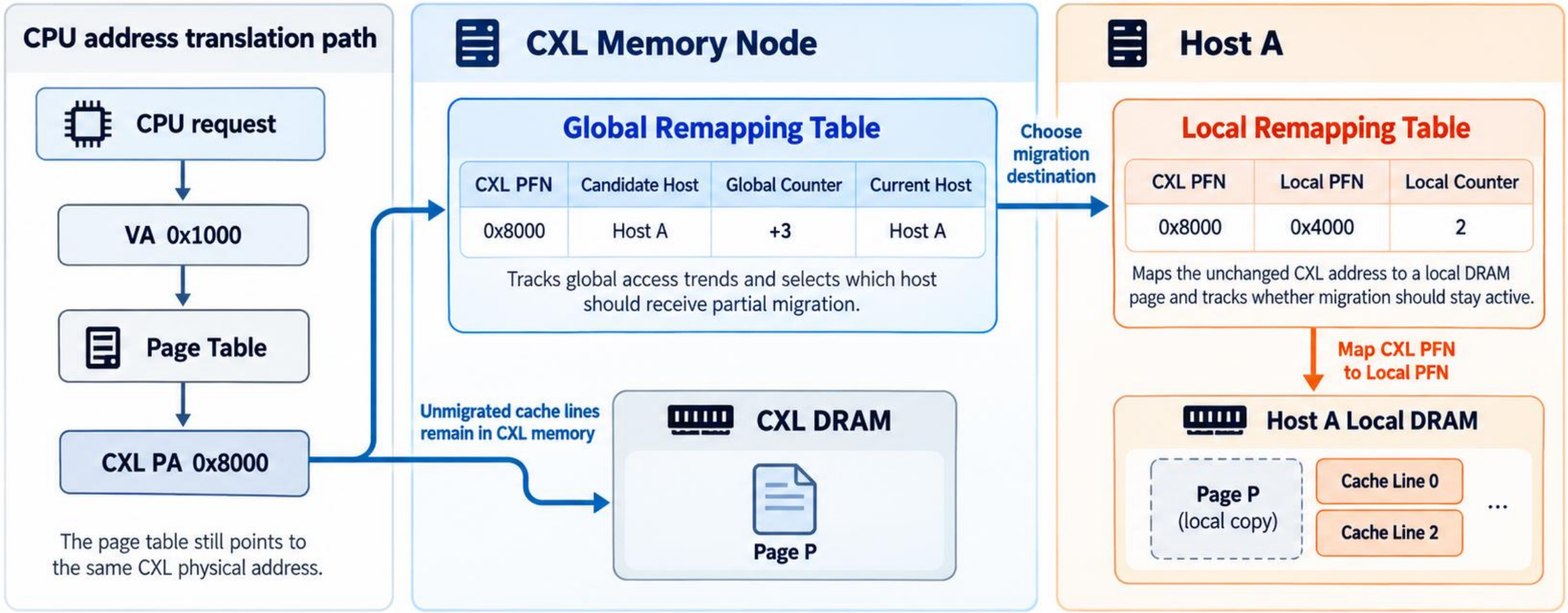
- Observation : Multi-host benefits from short migration intervals
  - management overhead, page transfer overhead becomes the dominant source of overhead

PIPM : Partial and Incremental Page Migration

Goals

- Reduce page transfer overhead
  - Granularity of migration : page -> cache line
- Reduce page management overhead
  - Global/local remapping table -> no need to update page table/inv tlb
- Solve local gain global pain problem
  - Majority-Vote migration polciy
- Design new coherence policy

Global and Local Remapping Tables in PIPM



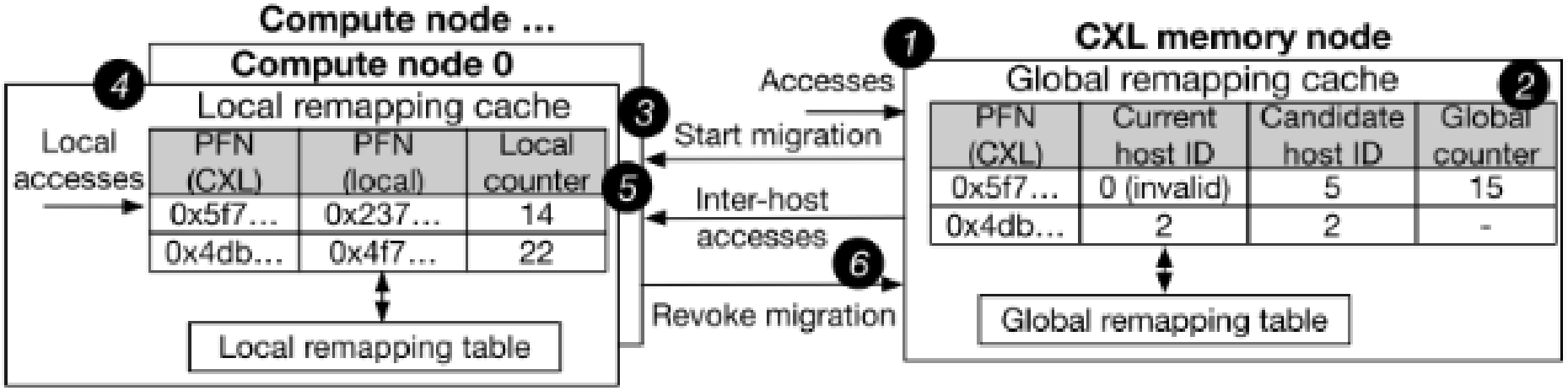
Summary

- **Global table** decides **WHERE** a page should migrate.
- **Local table** decides **HOW** the unchanged CXL address is served from local DRAM.

Majority-Vote Migration Policy

If $Access_i - \sum_{j \neq i} Access_j > threshold$, then migrate to host i

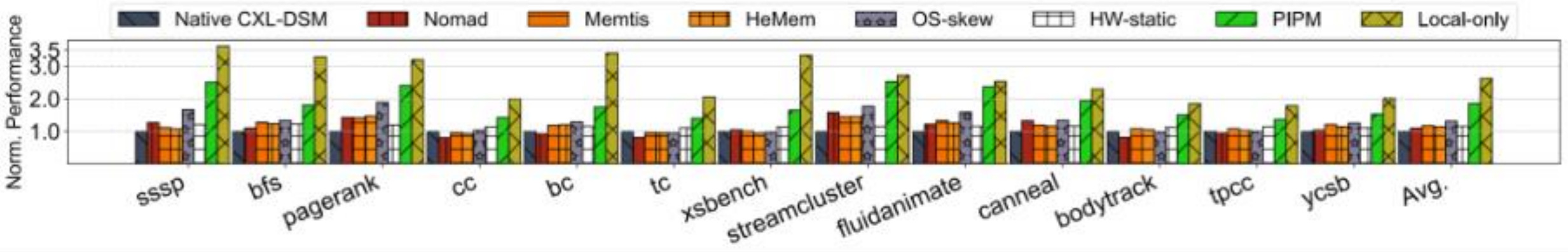
($Access_i$ == host i's number of access time to page)



Evaluation

End-to-end performance vs.

- Native CXL-DSM (no migration)
- Previous work (Nomad, Memtis, Hemem)
- OS-skew : only migration policy (is migration policy effective?)
- HW-static : only incremental migration(no migration policy) (is partial migration effective?)



Evaluation

- Average per-host memory allocation
    - OS-skew < prev
- Sensitivity to CXL Link Latency : relative performance improvement of PIPM over Native CXL-DSM under CXL-link latencies
- Sensitivity to area overhead : impact of global/local remapping table size
    - Negligible (local : 1MB, global : 16KB)